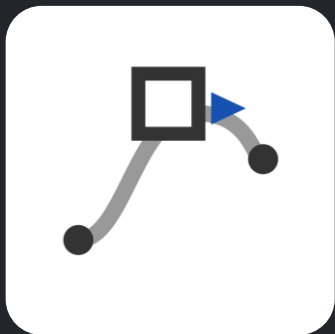




Move Along Edge

Move or copy bodies, mate connectors, sketch edges or points along an edge, chain or wire -- by a distance along it, or to the point nearest a target.



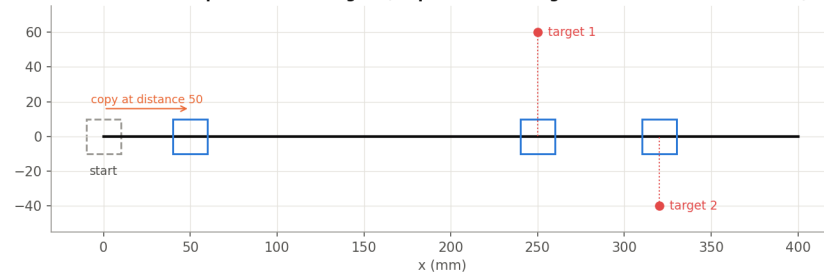
What it does

Native Transform cannot follow a curve by arc length, and Curve pattern spaces instances evenly. Move Along Edge moves (or copies) things a set distance along a path, or to where the path is nearest a target, turning them with the path if asked.

Use it when

- Place a part at a station along a curve
- Copy a detail to several named stations
- Put a point where a curve meets another

M15: main move to the point nearest target 1, copies nearest target 2 and at 50 mm -> x = 250, 320, 50

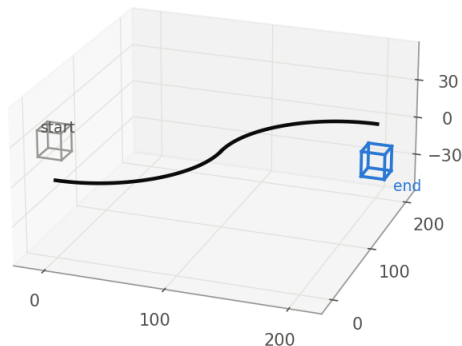


M15: moved nearest target 1; copies nearest target 2 and at 50 mm.

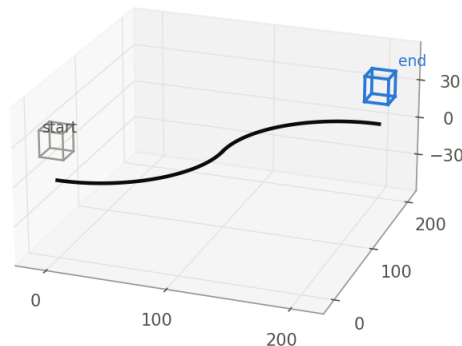


Orientation

Frenet (M7): follows the curvature normal -
flips at the inflection, the block ends below ($z = -30$)



Transported (M8): no twist -
the block stays on top ($z = +30$)



Tangent only

Default: the entity turns with the path's tangent.

Transported

No twist: a rotation-minimising frame (M8 stays on top).

Frenet

Follows the curvature normal: flips at inflections (M7 ends below), arbitrary on lines.



The dialog (1/2)



#	Parameter	What it means
1	Entities to move	Bodies, mate connectors, sketch edges, sketch vertices.
2	Edges or wire to move along	The path; positive = the first edge's own direction.
3	Provide Reference Point	+ vertex: use it instead of the centroid / connector origin.
4	Move to	Distance (signed, 25 mm; Flip direction or the arrow) / Nearest point to a target.
5	Apply	Move and rotate / Translate only / Rotate only.



The dialog (2/2)

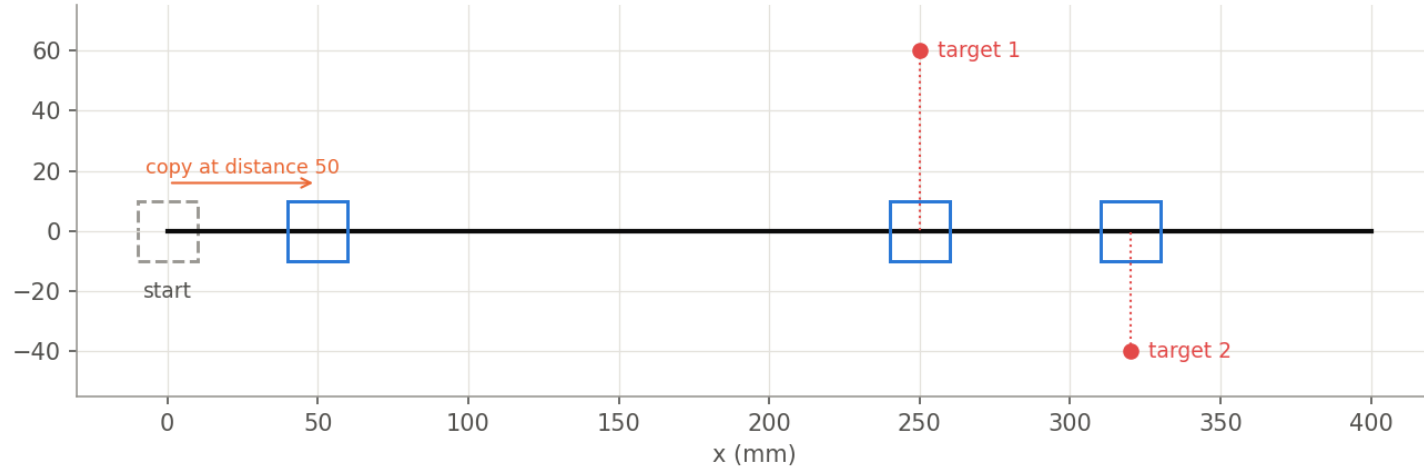


#	Parameter	What it means
6	Orientation	Tangent only / Transported (no twist) / Curve normal (Frenet).
7	Copy Bodies?	Copy instead of move.
8	Naming	New name / Prefix / Suffix + Name text.
9	Mate connectors as points	Output points instead of moving connectors.
10	Additional copies	More copies from the same start, each by distance or target, each named.



Example: nearest point and copies (M15)

M15: main move to the point nearest target 1, copies nearest target 2 and at 50 mm -> x = 250, 320, 50



Main move nearest (250, 60); copies nearest (320, -40) and at 50 mm

The cube lands at x = 250, copies at 320 and 50. Also tested: a closed loop wraps (M6), moving back past the start extends the path (M9), rotate only turns in place (M18).



Outputs and messages

Published for Extract variables

Key	What it holds
output / inputs	The moved entities (or copies) / the sources.
initial_copy	The first copy (Copy on).
copy_1 ... copy_n	The additional copies.

Messages

	Message / what to do
Info	rotate only on points / mate connectors cannot be named
Warning	source names missing Edit the feature once to refresh the names.



Tips and limits



Moves keep identity

A moved body is the same body: downstream references survive.



Nearest ignores Flip

And names refresh only on edit.



Frenet on lines

Undefined normal: use Transported.



Where it lives

Document	smallTools (14cdaad24a8beab67af40fe4)
Tab	Move_Along_Edge
Tests	Move along edge tests: M1-M20 + M3b (all pass)
Explainer	docs/explainers/small_tools/small_tools_explained.md, section 1
Related	Trim curve +; Extract variables (copy_n keys)